

# Matthew Van Winkle

Richmond, VA

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## EXPERIENCE

### **Research Assistant - Kai Larsen | Data Extraction and Pipelines | University of Colorado Boulder**

*October 2025 - September 2026, Boulder, Colorado*

- Built and owned an automated web + PDF data extraction pipeline processing 10K+ documents using Playwright, regex preprocessing, and LLM-based extraction.
- Transformed unstructured source data into normalized JSON datasets via fully scripted Python workflows with zero manual labeling for AI research consumption.
- Developed internal dashboards to monitor pipeline health and percentage coverage of available online data.

### **Course Assistant | University of Colorado Boulder**

*January 2024 - May 2025, Boulder, Colorado*

- Course Assistant for four semesters in Database Systems and Computer Systems; hosted office hours to reinforce applied SQL, relational modeling, and computer systems fundamentals.

### **Data Migration Intern | Allen, Allen, Allen, and Allen**

*May 2020 - August 2020, Richmond, Virginia*

- Supported migration from a legacy case management system to Litify (Salesforce-based CMS), assisting with data validation and workflow continuity.

## PROJECTS

### **Large-Scale Taxi Data Lakehouse & Analytics Platform**

- Engineered a 35M+ row ETL pipeline for NYC taxi data using Python, Polars, and DuckDB.
- Implemented a multi-layer architecture (Raw → Bronze → Silver → Gold → Platinum marts)
- Built analytics-ready marts powering executive dashboards
- Designed the system for batch processing with clear data contracts between layers.

### **Real-Time Market Data Pipeline with Latency Tracking**

- Built a real-time ingestion pipeline consuming live market data and powering a low-latency local dashboard.
- Implemented ingestion lag and end-to-end freshness tracking via timestamped logging and metrics tables.
- Designed a dual-path architecture: real-time operational cache and batched delivery to Snowflake for analytics.

## EDUCATION

### **Bachelor of Arts in Computer Science**

*August 2020 - July 2025*

- Technical GPA: 3.46
- Relevant Coursework: Database Systems, Data Visualization, Applied Machine Learning

## SKILLS

Languages: Python, SQL

Data: ETL pipelines, JSON transformation, data validation, schema mapping, batch & real-time ingestion.

Analytics and Visualization: Streamlit, Plotly, dashboard design, metric definition, Tableau.

Data Extraction: Web scraping, regex preprocessing, LLM-based extraction, OCR, cleaning.

Databases and Cloud: DuckDB, SQLite, Snowflake

Tools: Git/GitHub, APIs/REST, AWS (foundations), OpenAI tooling